

Supplementary information: Spatial association of
TLS programs and LAMP3⁺CCR7⁺ mregDC
states marks an immune-prognostic context in
ovarian cancer

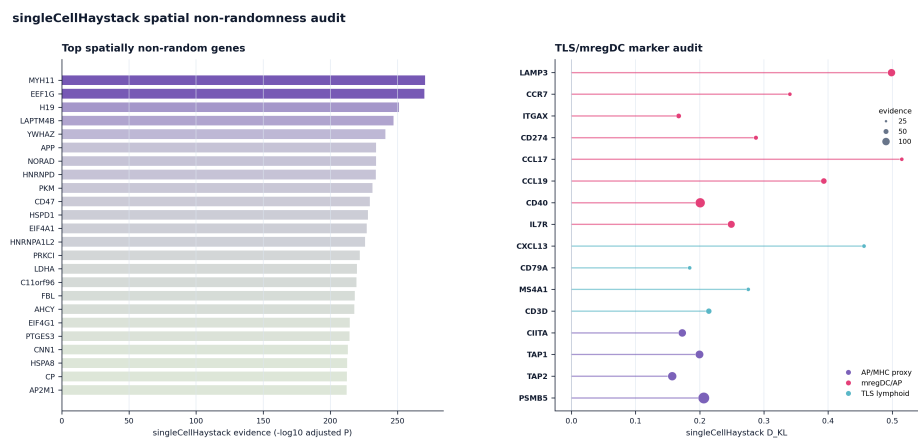
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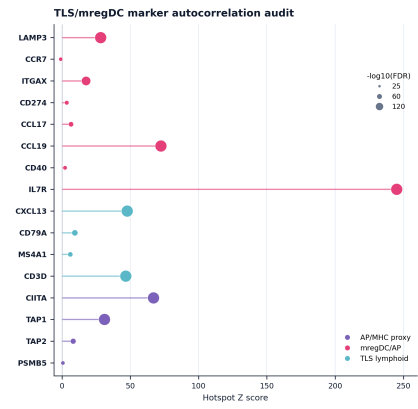
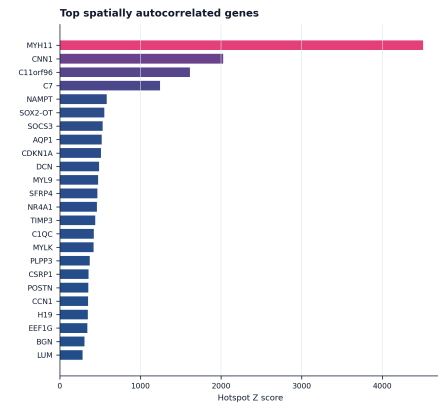
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Supplementary Figure Legends

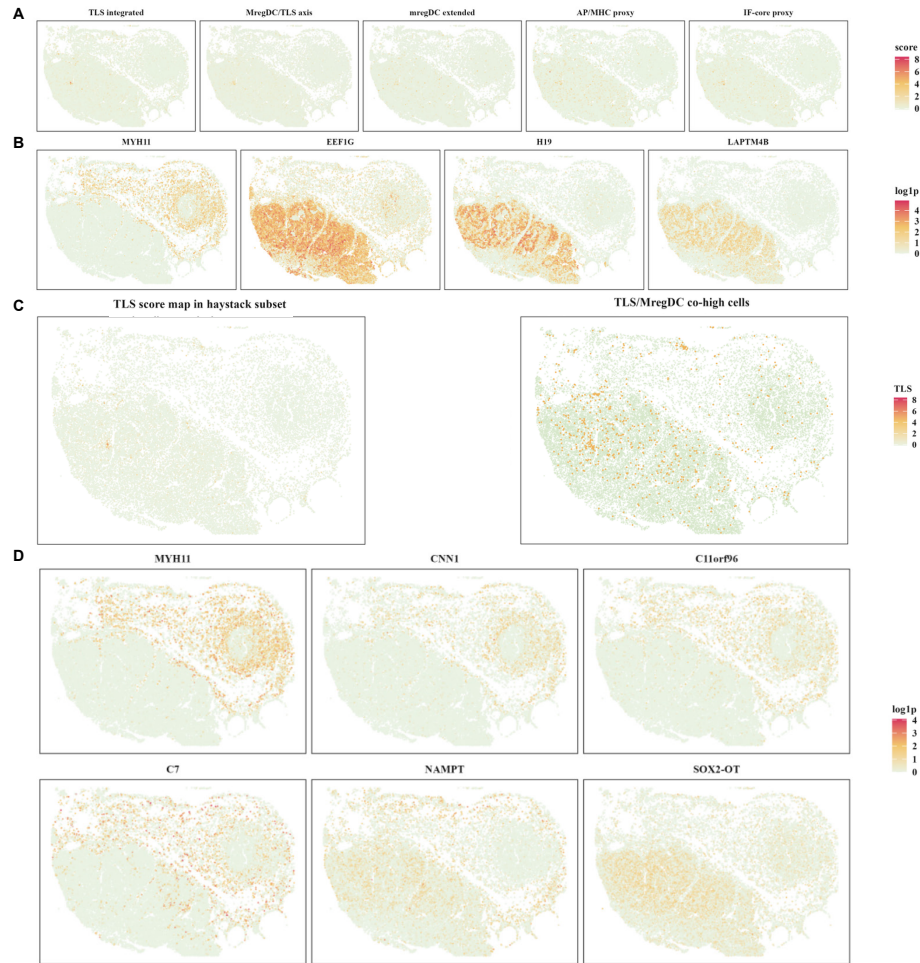


Supplementary Figure S1 singleCellHaystack spatial non-randomness audit. Ranked Xenium genes and marker-level evidence summarize spatially non-random transcriptional features in the deterministic ovarian cancer subset.

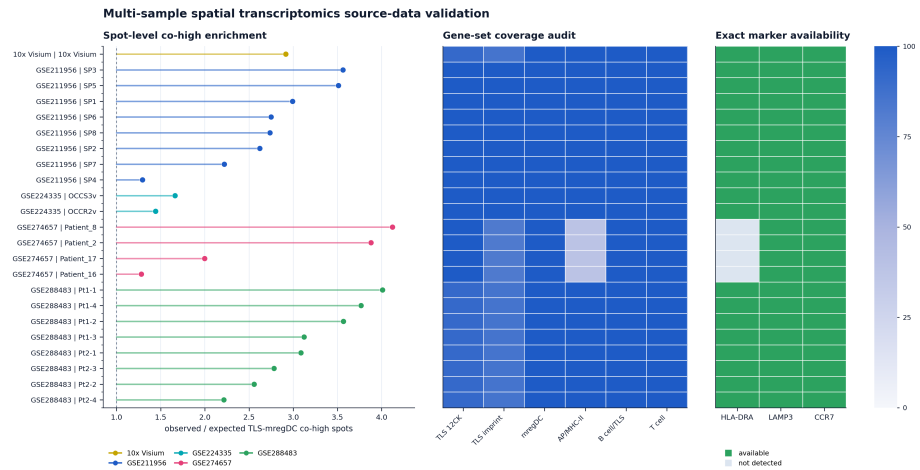
Hotspot spatial autocorrelation audit



Supplementary Figure S2 Hotspot spatial-autocorrelation audit. Ranked Xenium genes and marker-level Hotspot statistics summarize spatial autocorrelation of TLS/mregDC/antigen-presentation-associated features.

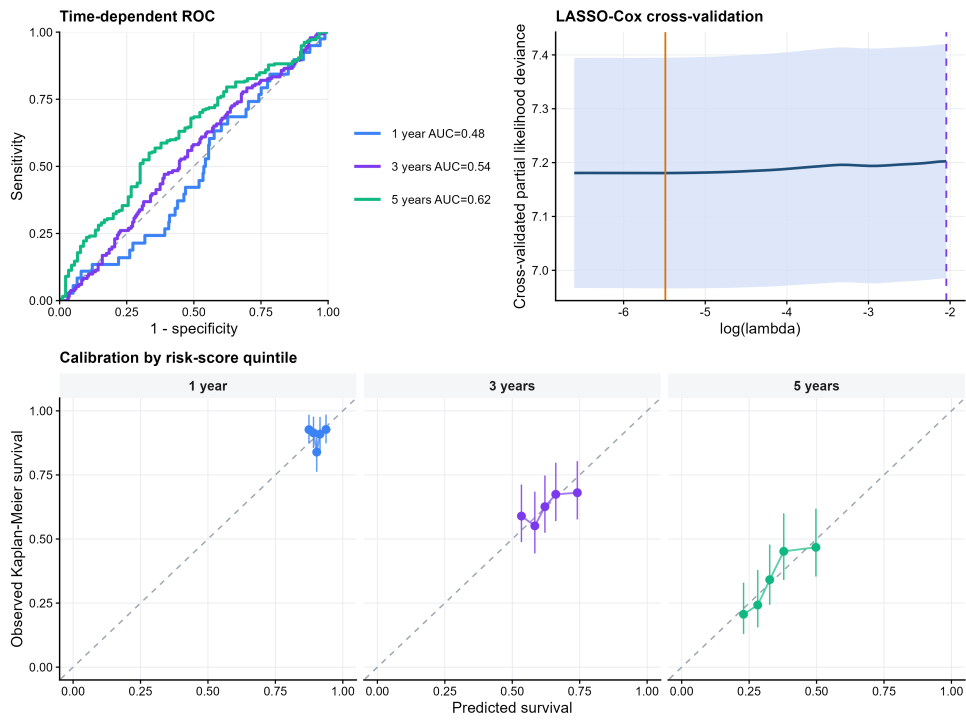


Supplementary Figure S3 Xenium spatial maps supporting feature audits. Module maps, non-random gene-expression maps, TLS-score alignment, co-high cells, and Hotspot representative genes are shown in the same spatial coordinate framework.

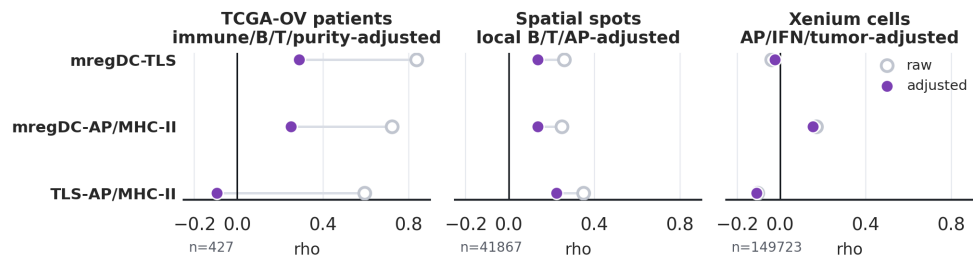


Supplementary Figure S4 Expanded multi-sample spatial source-data validation. Spot-level co-high enrichment, gene-set coverage, and exact-marker availability summarize public ovarian spatial datasets used for Fig. 5.

Exploratory prognostic model validation



Supplementary Figure S5 Exploratory model validation. Time-dependent ROC, LASSO-Cox cross-validation, and calibration-by-risk-quintile summaries support internal assessment of the TCGA-OV exploratory model.



Supplementary Figure S6 Immune-richness-adjusted specificity audit. Raw and adjusted Spearman correlations summarize TCGA-OV patient-level, multi-sample spatial, and Xenium sensitivity analyses for mregDC/TLS/AP associations.

Supplementary Source Data

- Current-numbering source-data map for main and supplementary figures: `source_data/SOURCE_DATA_INDEX.csv`.
- Fig. S1: `source_data/FigS01_singleCellHaystack_audit`.
- Fig. S2: `source_data/FigS02_Hotspot_audit`.
- Fig. S3: `source_data/FigS03_Xenium_spatial_maps`.
- Fig. S4: `source_data/FigS04_multi_sample_spatial_source`.
- Fig. S5: `source_data/FigS05_model_validation`.
- Fig. S6: `source_data/FigS06_specificity_adjustment`.
- Signature definitions, GEO source reconstruction, public spatial and 10x dataset accessions, and gene-coverage audits are provided in Supplementary Tables S1–S3 and `source_data/Tables_signature_and_GEO`.
- Legacy source-data folders are retained for traceability but should be interpreted through the current-numbering index.

Supplementary Tables

Supplementary Table S1 contains the reconstructed 22-sample scRNA-seq GEO source manifest, disease labels, sample counts, and accession-level summaries.

Supplementary Table S2 contains signature definitions, gene-coverage audits, marker-availability checks, and legacy-label harmonization for the mregDC, TLS, antigen-presentation/MHC-II, IFN, FDC/GC, HEV/stromal, proliferation, and EMT/invasion modules.

Supplementary Table S3 contains the 23 public multi-sample spatial transcriptomic samples used for Fig. 5/Fig. S4, including one public 10x Visium ovarian FFPE sample within the 23-sample spatial set, dataset-level GSE identifiers, sample-level GSM identifiers where applicable, and spot/cell counts. The same table also lists the separate public 10x scFFPE and Xenium Prime ovarian cancer reference datasets with URLs and software-version notes.